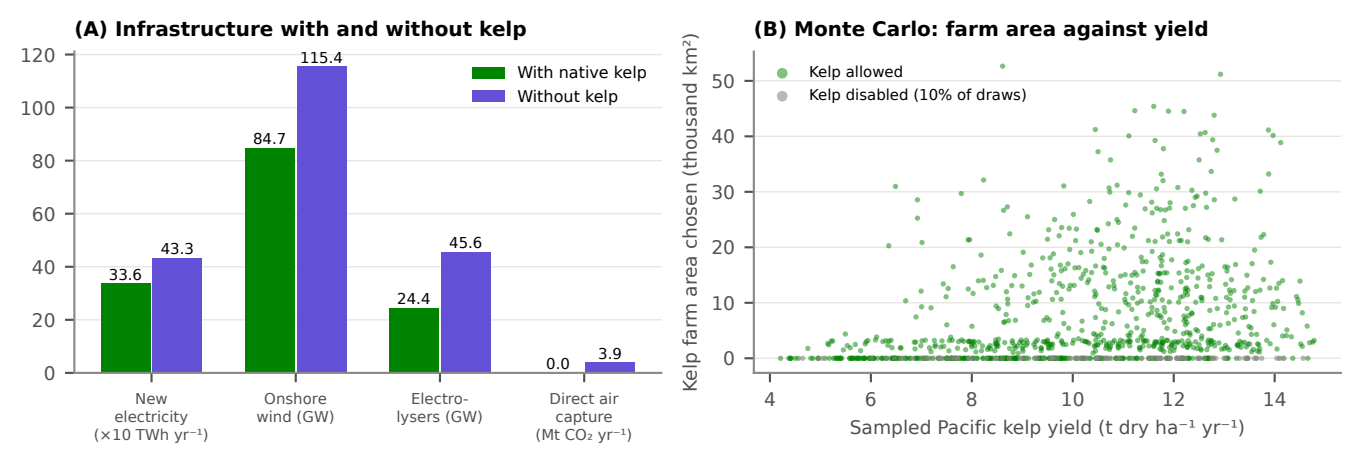
(A) Infrastructure with and without kelp
With native kelp
Without kelp
120
100
80
60
40
20
0
33.6
43.3
84.7
115.4
24.4
45.6
0.0
3.9
New electricity (×10 TWh yr⁻¹)
Onshore wind (GW)
Electro-lysers (GW)
Direct air capture (Mt CO₂ yr⁻¹)
(B) Monte Carlo: farm area against yield
Kelp allowed
Kelp disabled (10% of draws)
Kelp farm area chosen (thousand km²)
50
40
30
20
10
0
4
6
8
10
12
14
Sampled Pacific kelp yield (t dry ha⁻¹ yr⁻¹)